EşdeğerDers

Activities for the same learning objective as classroom resources change

Project summary. EşdeğerDers is a lesson preparation tool that generates three alternative activities, a shared assessment question and an answer key from the teacher's learning objective, student age, time limit, materials and technology access. Generative AI creates new teaching plans for changing conditions rather than merely embellishing text. A working prototype has been completed, with technical tests and internal content reviews. Learning impact has not yet been measured.

1 Problem target user and SDG connection

Our target user is a teacher preparing lessons for classrooms with limited materials or digital facilities. A single activity may become impractical when a screen, printer or specific object is unavailable. The United Nations SDG 4 framework highlights education access inequalities associated with income and location [1]. Our primary connection is **SDG 4 Quality Education**, particularly **Target 4.5**, which addresses equal access for vulnerable groups. The intended contribution is to offer alternatives that pursue the same learning objective using available resources; we have not measured a reduction in inequality.

2 Solution and working prototype

Teachers enter their classroom conditions, and activities are generated through short text/reading, hands-on practice and oral interaction. These are alternatives for the same objective, not consecutive lesson stages; each has its own time limit. The "This material is unavailable too" workflow removes selected materials and regenerates a plan with the same objective and remaining conditions. Previous and new plans can be compared. Teachers review resource warnings, objective alignment and the answer key, then print a student question sheet or a teacher plan containing answers.

Technology: React, TypeScript and Vite frontend; Python/FastAPI backend; Gemini API through the Google Gen AI SDK. The project's recorded model configuration is **gemini-3.8-flash** [2]. The key is kept only in the backend environment. POST /api/esdeger-ders/generate returns structured JSON. Three distinct methods, objective text and time limits are validated. Lack of classroom internet is an activity constraint; teachers still need connectivity for new AI generation.

3 Key prompts and design rationale

V1 is a short baseline instruction. V2 makes the role, resource constraints, children's rights, shared assessment and output contract explicit. The interface uses V2. The following are English translations of key excerpts from backend/prompts.py; the complete original prompts remain in the code [2].

V1 baseline system prompt
You are a teaching assistant. Prepare 3 different activities and 1 assessment question based on the given learning objective and classroom conditions.

V2 resource rule
You may use only the tools explicitly provided by the teacher in the Materials list.

V2 assessment and infeasibility instructions
Prepare a shared assessment question: it must assess the learning objective directly, not activity-specific details.
If a condition is infeasible or information is missing, write an explicit warning to the teacher in the uygulanabilirlik_notu field.

Prompt techniques and output validation

The user template receives {kazanim}, {yas_seviye}, {malzemeler}, {sure_dakika} and {erisim_kosullari}. **Role and context prompting** establishes the low-resource classroom focus. **Zero-shot instructions** define the task and methods; a fraction example in the system prompt serves as an accuracy anchor, rather than a few-shot arrangement of separate input-output demonstrations. **Step-by-step checking instructions** ask the model to review constraints, methods, timing and the shared question in sequence; a JSON schema constrains output structure. These instructions do not prove that the model's internal reasoning is correct. This report makes no measured superiority claim for V1 versus V2.

4 Tests results and challenges

Live end-to-end trial: Three activities were generated for age 9, a 10-minute limit per activity, paper/pencil/8 stones and no classroom digital access. After removing the stones, a separate live response changed the hands-on activity to paper folding; both plans were displayed in the interface. A separate noun-and-verb objective was also tested with live generation [2,3]. A transient service failure produced an explicit error; stored sample data was not presented as a live success.

Internal content review: Person 3 reviewed nine cases across mathematics, science, directions and Turkish: eight content generations and one empty-objective check. Findings included incomplete water-cycle explanations, limited coverage in a directions question and uncertain feasibility of two-minute activities [4]. Not all findings are considered resolved.

Technical evidence: The integrated frontend run passed 38 tests and the production build; student/teacher separation and page layout were checked in PDFs saved through Chrome [3]. In Person 4's separate QA snapshot, **73 of 81 automated tests passed and 8 failed**; the build passed. Thirteen live checks were blocked by a missing key, and 11 content/browser checks were not run [5]. Counts from different runs were not combined. These figures are not the model's accuracy rate.

Main challenges: Work addressed SDK JSON schema compatibility, clarifying time limits per alternative, safe API error presentation and matching material names. QA tests showed that controlled responses with incorrect resources or answers could pass the technical schema. Input size and API age/level validation remain improvements. Lexical resource warnings do not resolve every quantity, synonym or implicit requirement.

5 Ethical evaluation and human oversight

Key risks in child-facing content include misinformation, age-inappropriate instructions, unavailable resources and exclusion of low-resource classrooms. Prompts specify material and access constraints and prohibit clinical diagnoses, achievement labels and judgments of inadequacy. User text is sent separately from system instructions, with an anti-injection rule; actual attack resistance has not been established. Schema checks and resource scanning support teacher review; **they do not guarantee pedagogical accuracy or equal learning effects.**

There are no dedicated fields for student names, diagnoses or individual achievement records. However, personal information entered in free text may be sent to the API; unnecessary personal data should not be entered. Keys are excluded from the repository, and raw provider errors are not exposed to users. Session review checkboxes are not external expert approval. No external teacher interviews or classroom trials have been conducted.

6 Expected benefit and further evaluation

Next, teachers' normal preparation and tool-assisted preparation will be compared by recording time, required corrections, resource violations and perceived feasibility. Demo content should be reviewed again. The prototype's present contribution is generating traceable alternative plans for changing classroom resources; time savings, learning gains and SDG impact have not yet been demonstrated.

Sources and evidence

[1] UN SDG 4 and Target 4.5: sdgs.un.org/goals/goal4 (accessed 18 October 2026).
[2] backend/app.py, backend/prompts.py and docs/BACKEND_ENTEGRASYON.md; reviewed local revision: 451f542.
[3] docs/FRONTEND_TEST_RAPORU.md, 17 October 2026. [4] Person 3 educational content report, 17-18 October 2026.
[5] Kisi4_Guncel.zip: docs/TEST_RESULTS.md, BUG_REPORT.md, ETHICAL_EVALUATION.md and JUnit records, 18 October 2026.
Code and full prompts: github.com/ilaydameral/samsung-innovation-campus-esdeger-ders